On-Premises OpenShift Installation Considerations

When deploying **Red Hat OpenShift** on **on-premises infrastructure**, such as **VMware vSphere** or **bare metal**, several key factors must be considered to ensure a stable, high-performance, and secure environment.

1. Infrastructure & Hardware Requirements

CPU, Memory, and Storage Requirements

Ensure each node meets OpenShift's hardware requirements:

- ✓ **Masters:** Minimum 4 vCPUs, 16GB RAM
- ✓ **Workers:** Minimum 2 vCPUs, 8GB RAM (varies based on workload)
- ✓ **Bootstrap:** Minimum 4 vCPUs, 16GB RAM (temporary during installation)

Storage Considerations

- ✓ Choose a **high-performance, low-latency** storage backend (e.g., **vSAN, VMFS, NFS, iSCSI, Ceph, ODF**)
- ✓ Persistent storage must be available for applications and registry

Network Bandwidth & Latency

- ✓ Minimum **1 Gbps** for standard workloads; **10 Gbps** recommended for high-performance applications
- ✓ Ensure **low-latency networking** between master and worker nodes

DNS & NTP Configuration

- ✓ A **fully qualified domain name (FQDN)** is required for the OpenShift API and applications
- ✓ All nodes must use a **synchronized time source (NTP)** to avoid cluster instability

2. Deployment Platform-Specific Considerations

VMware vSphere Considerations

- ✓ Use **vSphere CSI Driver** for persistent storage integration
- ✓ Enable **vSphere DRS & HA** for workload availability
- ✓ Consider **VMware NSX-T or OpenShift SDN** for networking
- ✓ Allocate **static IPs or DHCP reservations** for OpenShift nodes

Bare Metal Considerations

- ✓ Use **Redfish, IPMI, or PXE boot** for automated provisioning
- ✓ Ensure **SR-IOV, GPU, or FPGA** support if needed for workloads
- ✓ Choose a **robust storage solution** like OpenShift Data Foundation (ODF)
- ✓ Validate **BIOS and firmware settings** for compatibility

3. Network & Security Considerations

Cluster Network Design

Define **CIDR ranges** for OpenShift components:

- ✓ **Cluster Network:** Default 10.128.0.0/14
- ✓ **Service Network:** Default 172.30.0.0/16
- ✓ **Ingress/Load Balancer** IP ranges

Load Balancer & Ingress

- ✓ Use an external **HAProxy, F5 BIG-IP, or MetalLB** for API and application routing
- ✓ Ensure **Load Balancer supports Layer 4 TCP forwarding** for OpenShift API

Security Hardening

- ✓ Enable **RBAC (Role-Based Access Control)** for user permissions
- ✓ Configure **NetworkPolicies** to restrict pod-to-pod communication
- ✓ Use **Advanced Cluster Security (ACS)** for vulnerability scanning

Firewall & Ports

- ✓ Open required ports for OpenShift services (e.g., **6443 (API), 22623 (MCS), 80/443 (Ingress)**)
- ✓ Restrict external access to only necessary endpoints

4. Installation Method Considerations

Installer-Provisioned Infrastructure (IPI) vs. User-Provisioned Infrastructure (UPI)

- ✓ **IPI (Automated):** OpenShift installer sets up everything (best for VMware)
- ✓ **UPI (Manual):** You control infrastructure provisioning (needed for bare metal)

Bootstrap Node Lifecycle

- ✓ Required **temporarily** for the OpenShift installation
- ✓ **Must be decommissioned after installation** for production clusters

Cluster High Availability (HA)

- ✓ Minimum **3 master nodes** for control plane HA
- ✓ **Worker node redundancy** to handle failures

5. Storage & Persistent Volume Considerations

Choose the Right Storage Backend

- ✓ **For VMware:** Use **vSphere CSI Driver, NFS, or OpenShift Data Foundation (ODF)**
- ✓ **For Bare Metal:** Use **Ceph, iSCSI, Fibre Channel, or local storage**

Storage Classes & Dynamic Provisioning

- ✓ Configure **StorageClasses** for dynamic volume provisioning
- ✓ Ensure **ReadWriteMany (RWX) support** if needed for shared storage

Registry & Logging Storage

- ✓ OpenShift **internal registry** requires persistent storage
- ✓ Logs should be stored in **Elasticsearch, Loki, or centralized storage**

6. Monitoring & Day 2 Operations

Enable OpenShift Monitoring

- ✓ Deploy **Prometheus, Grafana, and Alertmanager**
- ✓ Use **EFK (Elasticsearch-Fluentd-Kibana) or Loki** for log management

Backup & Disaster Recovery (DR) Strategy

- ✓ Use **Velero** or OpenShift Data Foundation (ODF) for backup & restore
- ✓ Implement **multi-cluster disaster recovery** with **Advanced Cluster Management (ACM)**

Upgrade & Lifecycle Management

- ✓ Plan for **rolling updates** of OpenShift and its components
- ✓ Use **OpenShift Cluster Version Operator (CVO)** for automated updates

To get the OpenShift Free downloads, please visit: https://assistedcloud.com/free-downloads/

7. Compliance & Governance Considerations

Identity & Access Management (IAM)

- ✓ Integrate OpenShift with **LDAP, Active Directory, or OAuth providers**
- ✓ Enforce **Multi-Factor Authentication (MFA) and RBAC policies**

Policy Enforcement

- ✓ Use **Gatekeeper (OPA)** or **Kyverno** for policy control
- ✓ Define security policies for **Pods, Namespaces, and Network Traffic**

Audit Logging & Compliance

- ✓ Enable **OpenShift Audit Logs** for tracking changes
- ✓ Ensure compliance with **CIS benchmarks, PCI-DSS, HIPAA, etc.**

Note:

When Deploying OpenShift on **VMware or bare metal** requires careful planning across **infrastructure, networking, security, storage, and operations**. Choosing the right deployment model (IPI vs. UPI), ensuring **HA, security, and monitoring**, and integrating with **storage and networking solutions** will ensure a **stable and scalable** OpenShift cluster.
